Lecture 3: Selecting a Solution

Remember: last time

- Data are diverse
  - Continuous vs discrete
  - Classification vs regression
  - Abundant vs rare (or asymmetric)
  - Multivariable
  - Available at once vs flowing
  - “Easily” modeled by statistical distributions

This time

- Machine learning solutions are based on these (and other) factors:
  - Eager vs lazy
  - Batch vs online
  - Parametric vs nonparametric
  - Discriminative vs generative

Lazy learners

- Instead of evaluating the training data, simply store it when it’s received

- When test data appear, classification happens using the subset of the training data that is most applicable to the test data

Eager learners

- Construct a model for classification when the training data is given, before seeing the test data

- Must be able to commit to a single hypothesis that can cover all data before seeing “all data” (the test data)

Eager learners

- Construct a model for classification when the training data is given, before seeing the test data
- Must be able to commit to a single hypothesis that can cover all data before seeing “all data” (the test data)
- Eager learners often take a long time to train and less time to predict

Batch (offline) learning

- “Standard” relative to online learning
  - Has been done for a much longer period of time
  - Is what most people would come up with if you described machine learning
  - Other X vs Y’s don’t have a “standard”

- Get a (training) dataset, build model with the dataset (which can be done in an eager or lazy way), then apply that model to another (test) dataset

Online learning

- Online learning processes data one-at-a-time, so you technically don’t need to store data (although you generally do)
- Online learning is adaptable because you don’t (can’t) make assumptions about the underlying distribution of the data
- Can react to changes in the data (for example, if consumers are responding to an event in the world)

When to use online learning

- Your data doesn’t fit into memory
- You expect your data to change over time (although there are ways to deal with this in batch learning, potentially obviously)
- Your data is a function of time and needs to be updated constantly (for example, stock prices)

Parametric learning

- Stays firm with the number of parameters needed to explain the data
  - No amount of training data will change the number of parameters
  - The training data *WILL* inform the coefficients of these parameters

$$b0 + b1*x1 + b2*x2 = 0$$

b0, b1, and b2 are coefficients that control the intercept and slope, x1 and x2 are variables

Parametric learning

- Can be bad or *really bad*
  - The previous example was modeling data as linear, but it may require a minor transformation to work out correctly
  - Alternatively, it could be modeling extremely non-linear data, which will likely cause the model to fail spectacularly

- Benefits of parametric learning:
  - Simple. Easy to understand and interpret
  - Fast. Can typically learn from data *very* quickly
  - Not data hungry. Can often learn quickly from little or incomplete data

Parametric learning

- Can be bad or *really bad*
  - The previous example was modeling data as linear, but it may require a minor transformation to work out correctly
  - Alternatively, it could be modeling extremely non-linear data, which will likely cause the model to fail spectacularly

- Limitations of parametric learning:
  - Constrained. By choosing a form, you're held fast to that form
  - Complexity. Only typically useful for low complexity problems
  - Poor fit. In practice, data rarely fits very well to underlying model

Nonparametric learning

- *Do not* make strong assumptions about the form of the mapping function

- Seek the best fit of the training data while maintaining the ability to generalize in case of unseen data

- “Nonparametric methods are good when you have a lot of data and no prior knowledge, and when you don’t want to worry too much about choosing just the right features.” (Artificial Intelligence: A Modern Approach, page 757)

Nonparametric learning

- Benefits:
  - Flexible
  - Power (no assumptions make for powerful methods)
  - Performance (can “fit” very well)

- Drawbacks:
  - Require a good amount of data
  - Slow (both per unit data and because they require more data)
  - Overfitting (we’ll talk about this more later, but nonparametric methods are more susceptible to overfitting, which is bad)

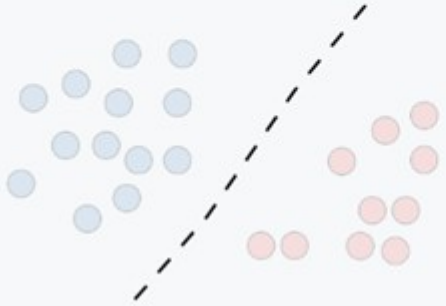
Discriminative vs generative learning -- analogies

- You want to figure out what language someone is speaking:
  - Generative – learn every language, determine which language the person is speaking by figuring out which one it belongs to
  - Discriminative – determine linguistic differences without learning any languages

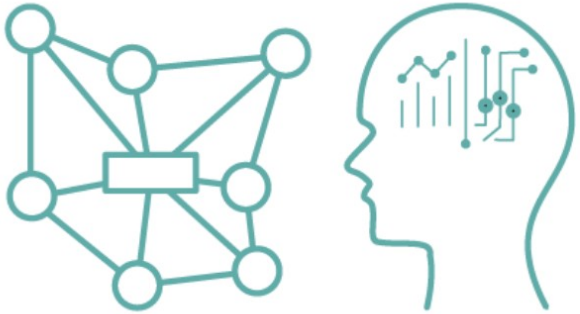
- 2 kids: 1 (“kid A”) learns by memorizing everything in depth, the other (“kid B”) learns the differences between what is seen. Both go to the zoo
  - “A” identifies a photo of a lion based on the **closest match** to the zoo animals
  - “B” identifies the same photo as a lion based on observed **characteristics** of lions

Discriminative vs generative learning

- To put it another way,
  - You can put together a model for what, say lions and tigers, look like, then you create a boundary that separates them. You can decide whether the current example falls in with the lions of the tigers (discriminative)
  - You could also build a model for tigers, build a model for lions, then compare the current “test data” point against both the tiger model and lion model and find which it behaves more like (generative)

	Discriminative model	Generative model
Goal	Directly estimate $P(y\|x)$	Estimate $P(x\|y)$ to then deduce $P(y\|x)$
What's learned	Decision boundary	Probability distributions of the data
Illustration		

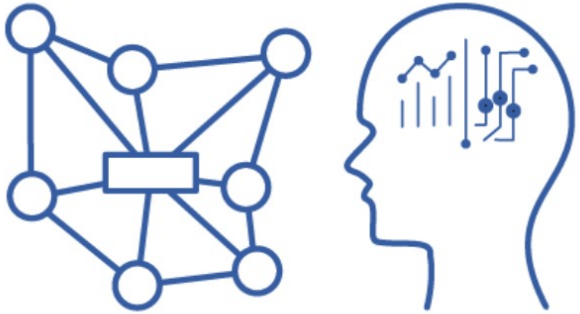
Additional consideration



Supervised

Classification

Regression



Unsupervised

Clustering

Collaborative Filtering

Frequent Pattern Mining